Based on the work of Adamowicz & Bartlett (1987):

Optimized virtual orbital space for high-level correlated calculations

Adamowicz, L. & Bartlett, R. J. [*J. Chem. Phys.* **86**, 6314-6324 (1987)]

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Reduce the virtual orbital space dimension from N_{ACT} to $N'_{ACT} < N_{ACT}$ while preserving most correlation energy.

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Key Equations

The second-order Hylleraas functional:

$$E_2 \leq \langle \Phi_1 | H_0 - E_0 | \Phi_1 \rangle + 2 \langle \Phi_1 | V - E_1 | \Phi_0 \rangle = J_2$$

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MP2 correlation energy expression:

Key Equations

Gradient and Hessian expressions:

$$G_{EA} = \frac{\partial J_2}{\partial R_{EA}} = \underbrace{2 \sum_{I>J} \sum_{B} t_{IJ}^{AB} \langle IJ || EB \rangle}_{\text{two-electron response}} + \underbrace{2 \sum_{B} D_{AB} f_{EB}}_{\text{Fock response}}$$

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$$H_{EA,FB} = \frac{\partial J_2}{\partial R_{EA} R_{FB}} = 2 \sum_{I>J} t_{IJ}^{AB} \langle IJ || EF \rangle + - \sum_{I>J} \sum_{C} \left[t_{IJ}^{AC} \langle IJ || BC \rangle + t_{IJ}^{CB} \langle IJ || CA \rangle \right] \delta_{EF}$$

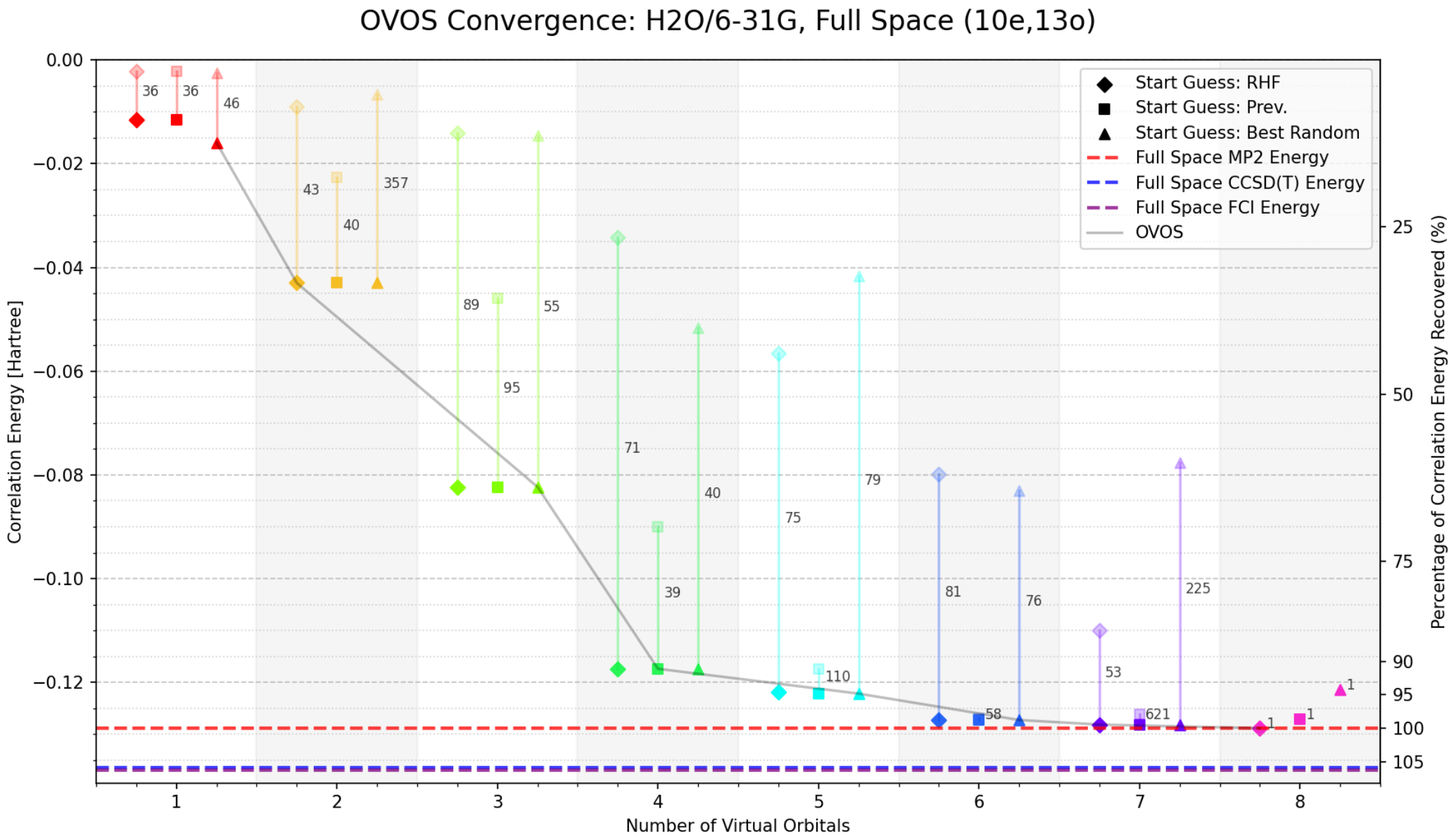
Key Equations

Newton-Raphson update for orbital rotations:

$$\mathbf{R_{EA}} = -\mathbf{G}_{EA} \cdot \mathbf{H}^{-1}_{EA,FB}, \qquad \mathbf{R} = \begin{bmatrix} \mathbf{0} & -\mathbf{R_{EA}} \\ \mathbf{R_{AE}} & \mathbf{0} \end{bmatrix}$$

define the orbital

Optimized Virtual Orbital Space (OVOS)



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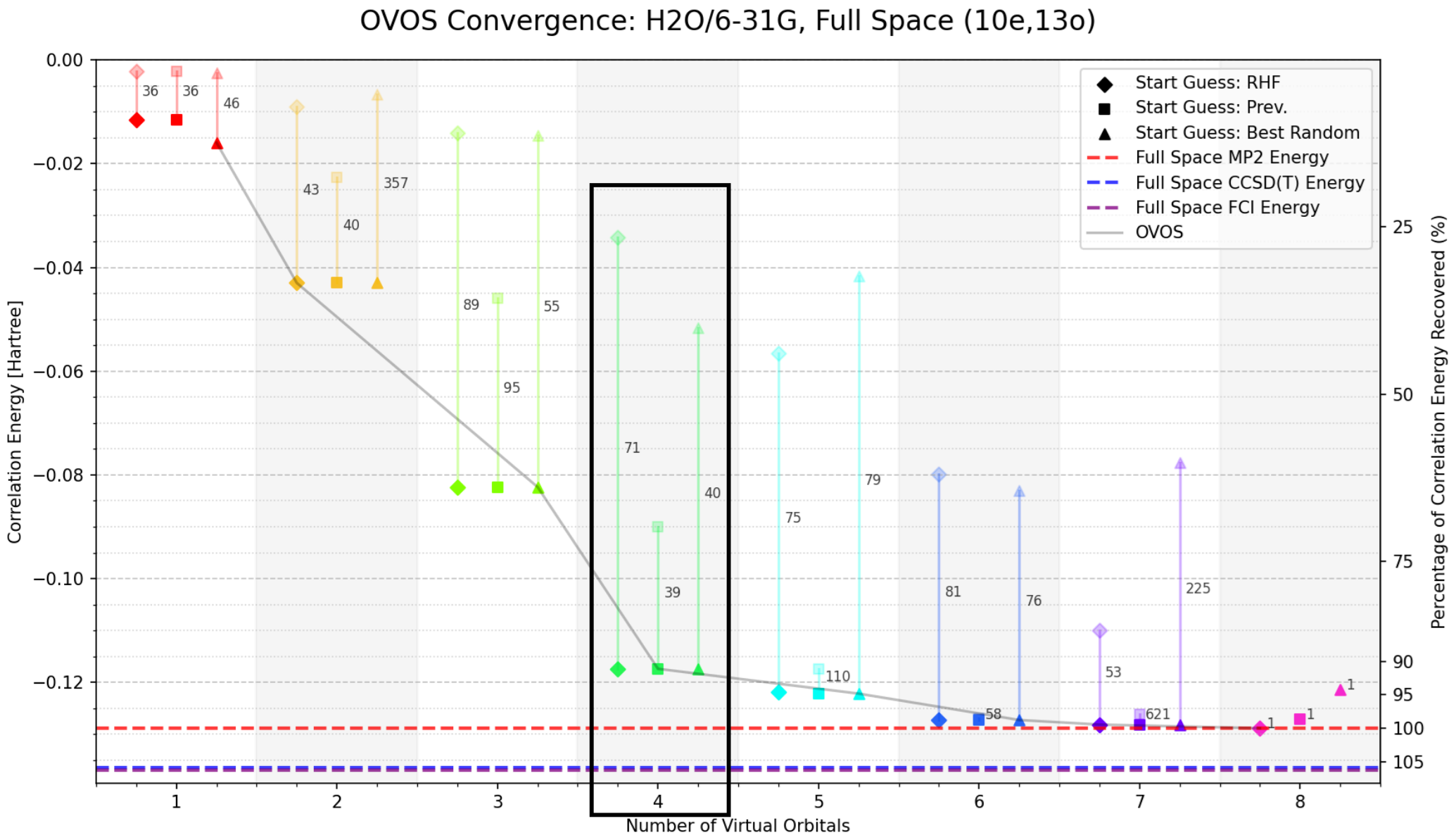

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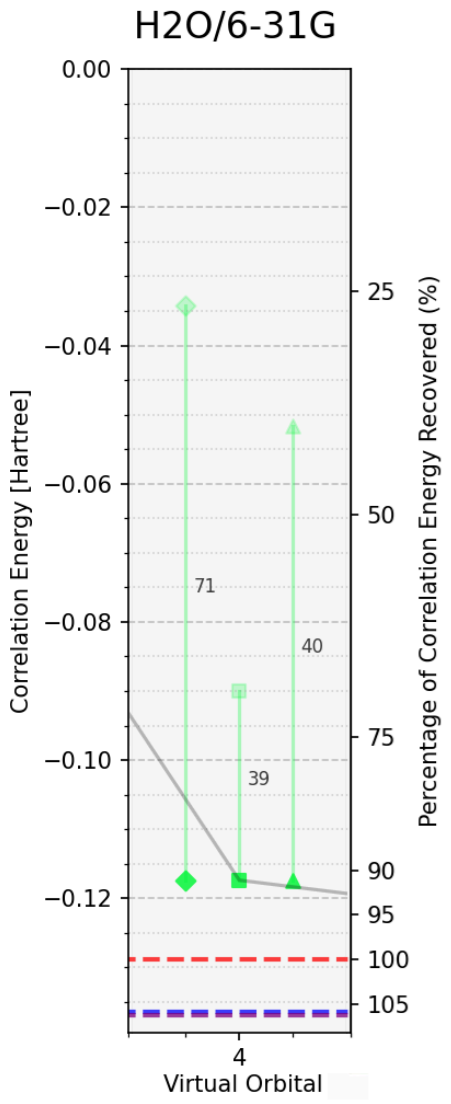
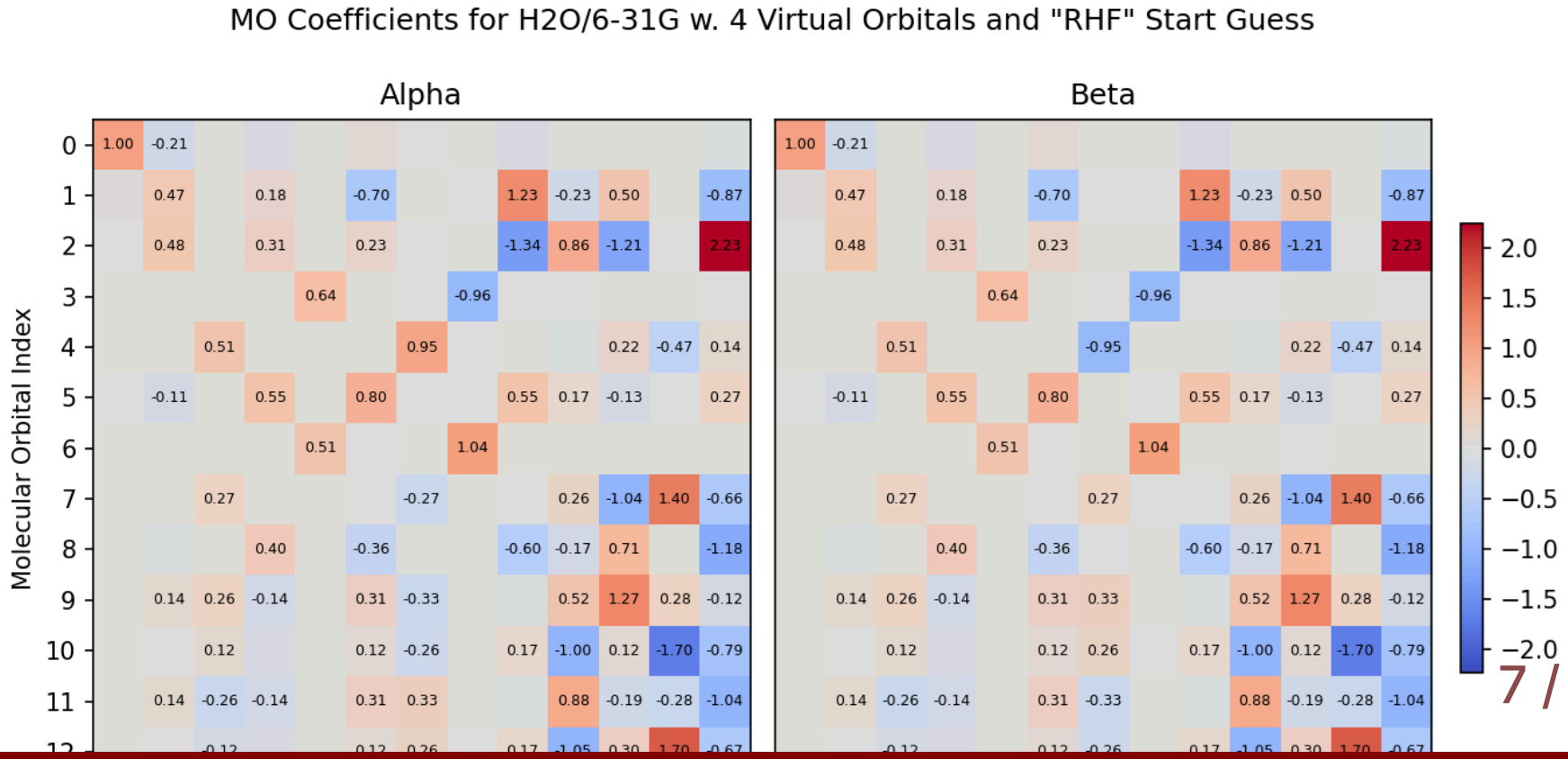


Molecular orbital coefficients



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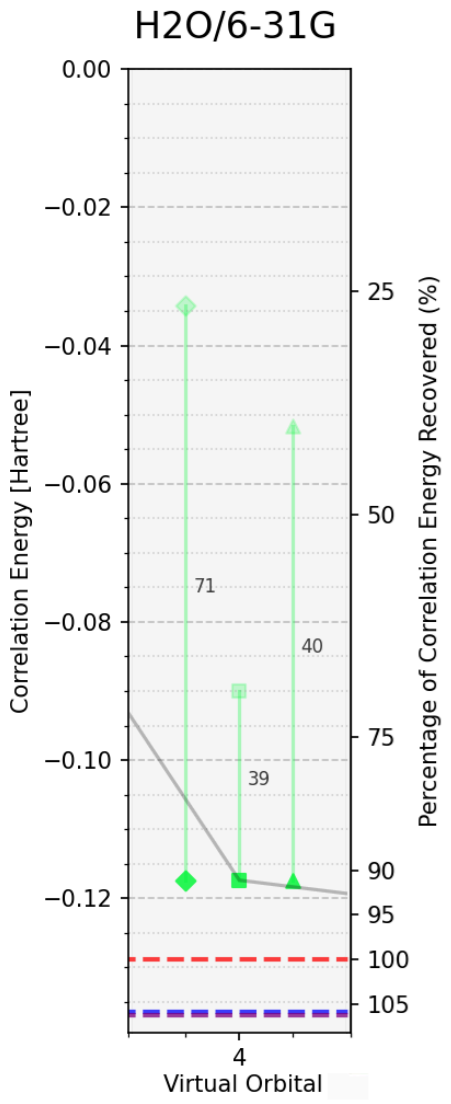

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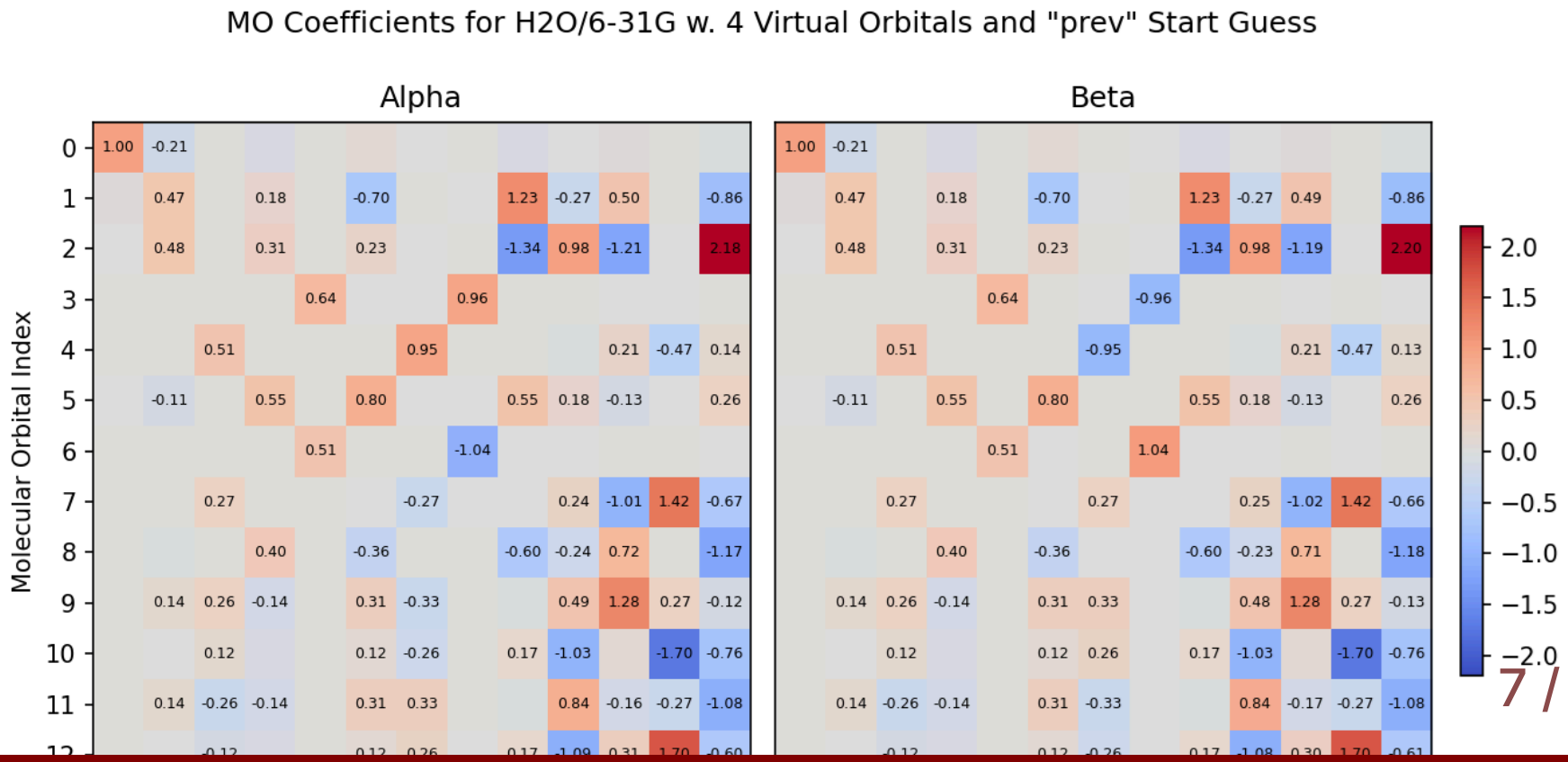

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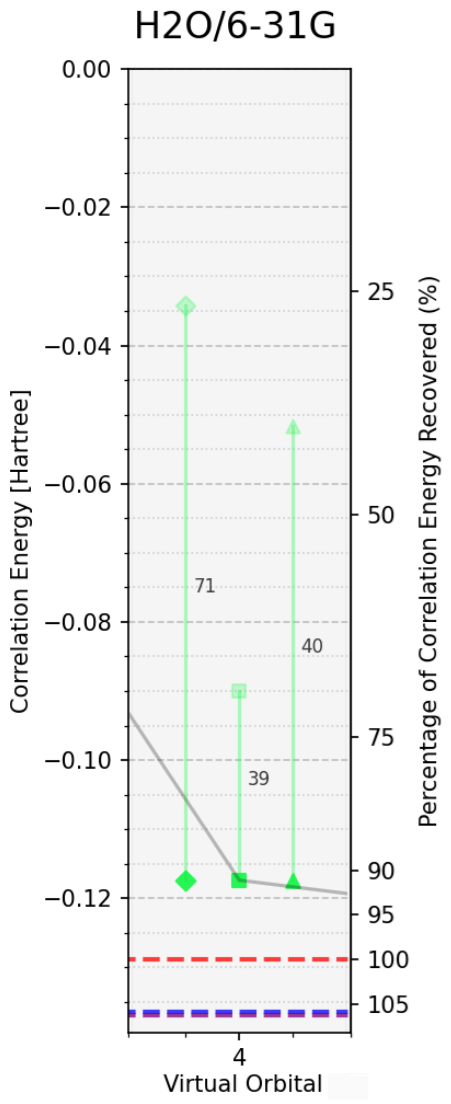
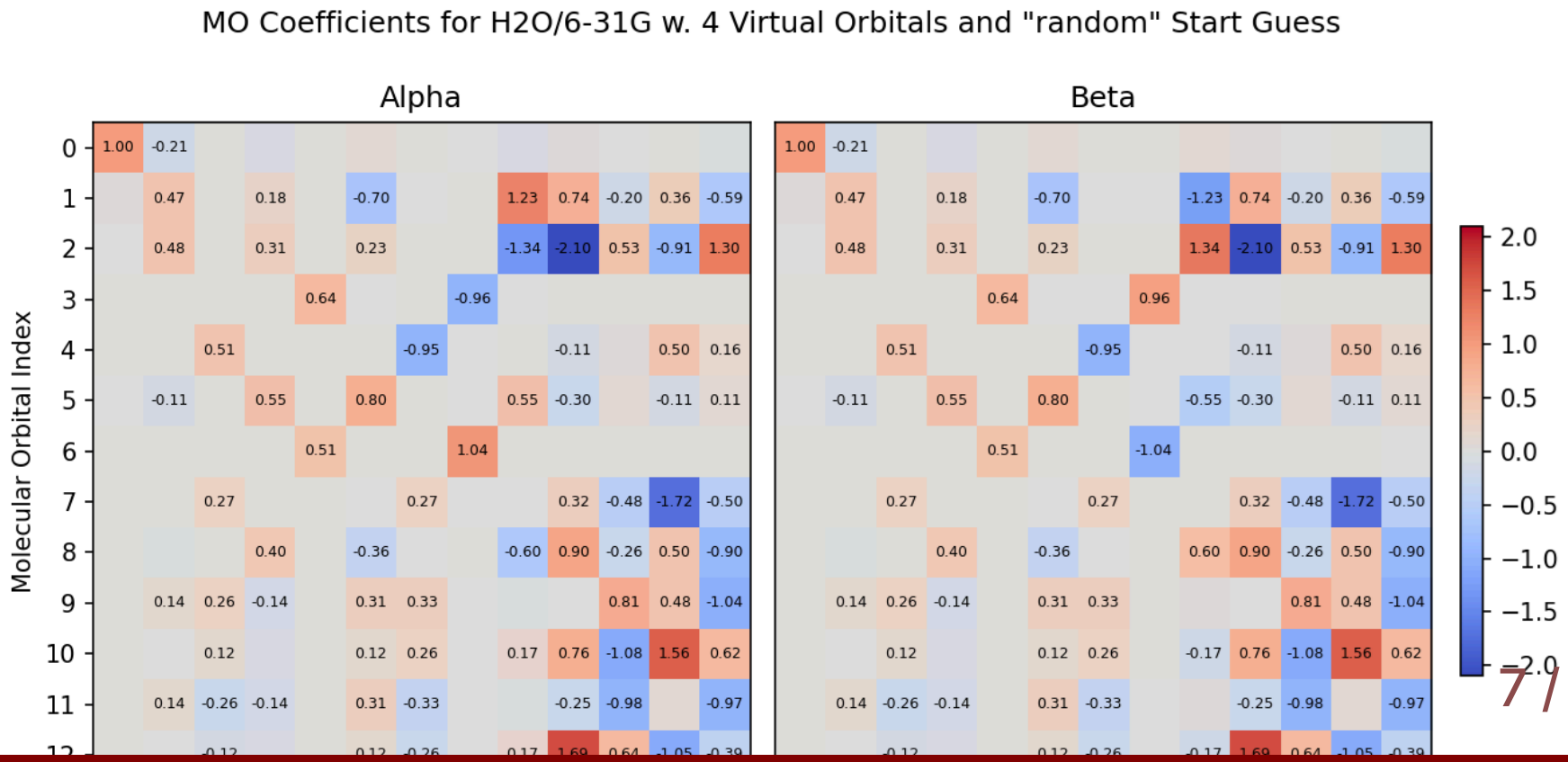
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